Experiment 5: Document Question Answering Using Retrieval-Augmented Generation (RAG)

Aim

To develop a document question-answering system using Retrieval-Augmented Generation, combining document retrieval with a language model.

Algorithm

1. Create a collection of documents.
2. Generate embeddings for the documents.
3. Retrieve the most relevant document for a question.
4. Extract the retrieved document as context.
5. Pass the context and question to a language model.
6. Generate an answer based on the retrieved context.
7. Display the answer.

Code

```
# Experiment 5: Basic RAG Question Answering


documents = [
    "Artificial intelligence enables machines to perform tasks that normally require human intelligence.",
    "Machine learning is a branch of AI that learns patterns from data.",
    "Natural language processing enables computers to process and understand human language.",
    "Computer vision enables computers to analyze images and videos."
]


# Generate embeddings
doc_embeddings = embedding_model.encode(
    documents,
    normalize_embeddings=True
)
```

```
question = input("Ask a question: ")

# Retrieve the most relevant document
query_embedding = embedding_model.encode(
    [question],
    normalize_embeddings=True
)[0]

scores = np.dot(doc_embeddings, query_embedding)
best_index = np.argmax(scores)

context = documents[best_index]

print("\nRetrieved Context:")
print(context)

# Generate answer using the local language model
answer = generate_answer(context, question)

print("\nGenerated Answer:")
print(answer)
```

Sample Input

Ask a question: What is machine learning?

Sample Output

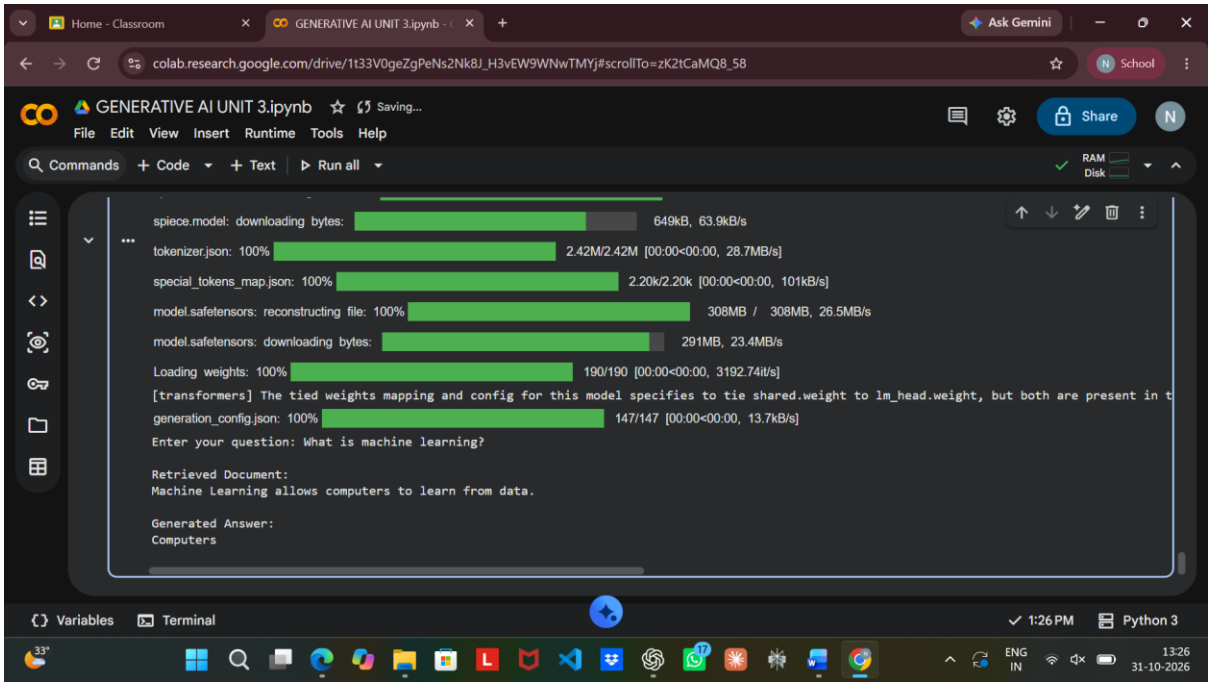
Retrieved Context:

Machine learning is a branch of AI that learns patterns from data.

Generated Answer:

Machine learning is a branch of AI that learns patterns from data.

The generated answer may be worded differently.



Result

A basic RAG system was implemented by retrieving relevant document context and providing it to a local language model for answer generation.